

Question number	Answer	Additional guidance	Mark
1(a)	B is the correct answer A is not the correct answer glycolysis, Krebs cycle and link reaction is not the correct order. C is not the correct answer as link reaction, Krebs cycle and glycolysis is not the correct order. D is not the correct answer as Krebs cycle, glycolysis and link reaction is not the correct order.		(1)

Question number	Answer	Additional guidance	Mark
1(b)	An answer showing the following steps: - equation (1) - balanced equation (1)	accept incorrect formula e.g. CO₂ accept correct word equation e.g. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ ignore ATP / ADP 1 mark if equation correct but not balanced	(2)

Question number	Answer	Additional guidance	Mark
1(c)(i)	B is the correct answer A is not the correct answer as 0.67 is not the RQ value C is not the correct answer as 1.06 is not the RQ value D is not the correct answer as 1.42 is not the RQ value		(1)

Question number	Answer	Additional guidance	Mark
1(c)(ii)	A calculation showing the following steps: - calculation of energy in ATP (1) - calculation of percentage useable energy with answer given to three significant figures (1)	$(122 \times 30.51) = 3722.22 \text{ kJ}$ $(122 \div 12244) = 0.009964$ $(3722.22 \div 12244) \times 100 = 30.4\%$ $(0.009964 \times 30.51) \times 100 = 30.4\%$ Correct answer gains both marks	(2)

